

# Topology First-Year Exam, January 2024, Part 1

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*For the first five problems, show all your work and support all statements.*

1. Prove that  $\{0, 1\}^\omega$  is not countable.
2. Show that  $\mathbb{R}^\omega$  with the box topology is not connected.
3. Assume that  $f : X \rightarrow Y$  is continuous and surjective.
  - (a) If  $X$  is Lindelöf (every open covering has a countable subcovering), show that  $Y$  is also.
  - (b) If  $X$  is separable (has a countable dense subset), show that  $Y$  is also.
4. Let  $(X, d)$  be a compact metric space. Let  $f : X \rightarrow X$  and suppose there is some  $c < 1$  with  $d(f(x), f(y)) \leq c \cdot d(x, y)$  for all  $x, y \in X$ . Prove there exists a unique  $x \in X$  with  $f(x) = x$ .
5. Let  $(Y, d)$  be a complete metric space. Let  $J$  be a set. Suppose  $d(y, y') \leq 1$  for all  $y, y' \in Y$ , meaning a simpler definition for the uniform metric  $\bar{\rho}$  on  $Y^J$  is

$$\bar{\rho}(f, g) = \sup_{\alpha \in J} \{d(f(\alpha), g(\alpha))\}$$

for  $f, g : J \rightarrow Y$ . Prove that  $(Y^J, \bar{\rho})$  is a complete metric space.

*Answer the following problems with complete definitions, complete statements, an example, or a short proof.*

6. Define what it means for a topological space  $X$  to be regular.
7. Show that a closed subspace  $C$  of a compact space  $X$  is compact.
8. A function  $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$  is eventually zero if there is a positive integer  $N$  such that  $f(n) = 0$  for all  $n \geq N$ . Prove that the set of all functions  $f : \mathbb{Z}_+ \rightarrow \{0, 1\}$  that are eventually zero is countable.
9. Show that  $[0, \infty)$  and  $\mathbb{R}$  are not homeomorphic.
10. Let  $x = (x_1, x_2, \dots) \in \mathbb{R}^\omega$ . Let  $\{x^{(n)}\}_{n \in \mathbb{Z}_+}$  be a sequence of points in  $\mathbb{R}^\omega$  such that for each coordinate  $i \in \mathbb{Z}_+$ , the sequence  $\{x_i^{(n)}\}_{n \in \mathbb{Z}_+}$  converges to  $x_i$  in  $\mathbb{R}$ . Show that  $\{x^{(n)}\}_{n \in \mathbb{Z}_+}$  converges to  $x$  if  $\mathbb{R}^\omega$  has the product topology.
11. Give an example of a topological space that is connected but not path-connected.
12. Define what it means for a topological space  $X$  to be a Baire space.
13. Show that if a topological space  $X$  is Hausdorff, then a sequence in  $X$  can converge to at most one point in  $X$ .
14. Show that the function  $f : [0, 2\pi) \rightarrow \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$  from the half-open interval to the circle defined by  $f(t) = (\cos(t), \sin(t))$  is not a homeomorphism.
15. Let  $X$  be a locally compact Hausdorff space. Define the topology on  $Y = X \cup \{\infty\}$  that makes  $Y$  the one-point compactification of  $X$ .